TUTORIAL ON HUMAN FACTORS IN CROWDSOURCING

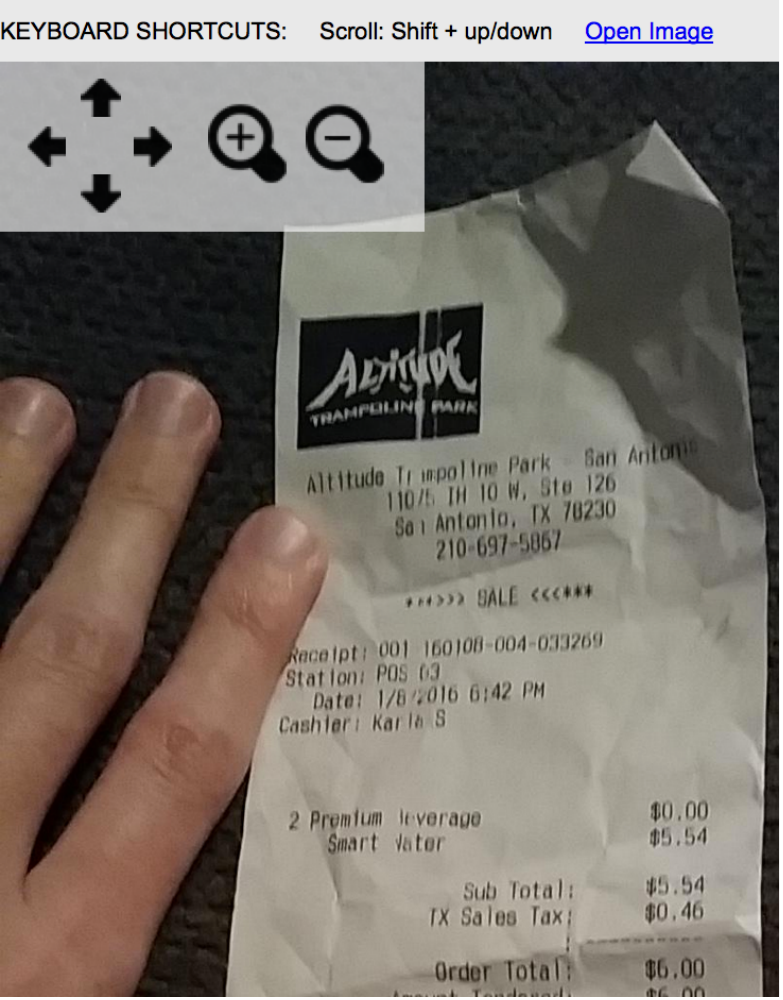
Sihem Amer-Yahia

Senjuti Basu Roy

Crowdsourcing Today

- A place where one finds work, possibly with *remuneration.*
- Uber-ization of human labor is a big phenomenon today.
- *Micro-tasks* often easier to complete by humans than by machines.
- Crowd is *volatile*, its arrival and departure *asynchronous*, its levels of attention, motivation, and accuracy *diverse*.

A common (micro)-task in AMT



Classify Receipt

Hit Reward: $0.02

Real readable original receipt | Not a receipt or not readable

The following details can often be found at the top or bottom of the receipt. Enter as much information as you can find.

Find and enter the business phone number:

Phone

Example: (888) 555-1234 or 8885551234

Find and enter the business address:

Address

City

State | Postal code

Example: 321 Fake Street, Los Angeles, CA, 90210

Next

CrowdMap from Ushahidi

ALL CATEGORIES

DAMAGE IMAGES

AID IMAGES (REPORTED DURING DAMAGE SEARCH)

TRUSTED REPORTS

OTHER LAYERS [HIDE]

TYPHOON TRAJECTORY

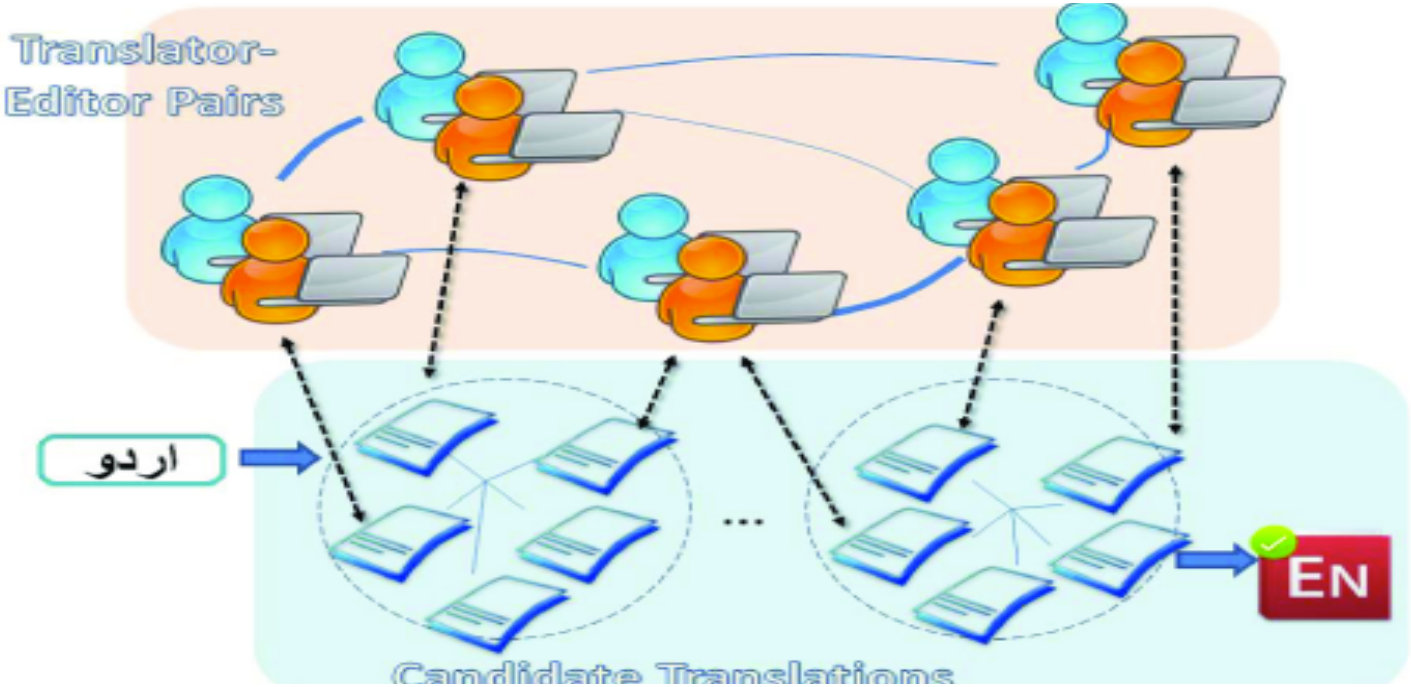
TOMNOD CROWDRANK

HOW TO REPORT

By using an app:
iPhone
Android

By sending an email:

Urdu-to-English Translation



Are Two Heads Better than One? Crowdsourced Translation via a Two-Step Collaboration of Non-Professional Translators and Editors. Rui Yan, Mingkun Gao, Ellie Pavlick, and Chris Callison-Burch

Many more examples

- Micro-tasks
  - audio transcription, text translation, image tagging, citizen science
  - implicit collaboration
  - consensus usually achieved with majority voting

- Collaborative tasks
  - a group of individuals *collectively working* to achieve a goal
  - collaborative editing, fan-subbing, solution outsourcing (e.g., Netflix contest)
  - Consensus achieved when crowd converges

This tutorial's goal

- Reliance on workers and ***human factors*** (characteristics of human beings and their environment) is both a challenge and an opportunity.
- Classic motivation theory, organization studies, and open-source software development, understand *human factors,* related to aspects of work *in physical workplaces*.
- Attempt ***to illustrate and formalize human factors in virtual marketplaces*** by drawing lessons from the social sciences.

Outline

- PART 1: Human Factors at Work
- PART 2: Human Factors in Task Assignment and Task Completion
- PART 3: Evaluation and Future Directions

9

PART 1
Human Factors at Work

Humans Factors, a rough characterization

- Worker-specific
  - Micro-tasks: Skill, Reputation/Trust, Expected Pay, Acceptance Ratio
  - Collaborative tasks: Affinity, Critical Mass, Interaction model
- Task-specific
  - Feedback, Incentives, Expected Quality, Budget, Desired Expertise
- Workers and tasks
  - Motivation

Human Factors in practice

- Acquiring/learning human factors
  - Qualification tests
  - Assigning tasks to workers to derive their factors (e.g., skill)
  - Asking workers (e.g., expected wage)
- Leveraging human factors
  - In task assignment: self-appointment or automatic assignment
  - In task completion: feedback and motivation to complete tasks

Worker-Specific: Skill and Trust

paper	Type of task	Contributions
Whose vote should count more: Optimal integration of labels from labelers of unknown expertise. ***Whitehill et al.*** *NIPS 2009*	Micro-task	A probabilistic model to infer the true label of each image, the expertise of each labeler, and the difficulty level of each task.
Supervised learning from multiple experts: whom to trust when everyone lies a bit. ***Raykar et al.*** *ICML 2009*	Micro-task	A Bayesian framework that iteratively establishes and refines a particular gold standard, and measures the performance of the annotators wrt that standard.
Eliminating spammers and ranking annotators for crowdsourced labeling tasks. ***Raykar et al.*** *JMLR 2012*	Micro-task	A Bayesian algorithm that iteratively eliminates spammers by ranking annotators.
Evaluating the crowd with confidence. ***Joglekar et al.*** *KDD 2013*	Micro-task	Determines confidence intervals and worker error rates that characterize the range in which worker errors lie.

Worker-Specific: Skill and Trust

paper	Type of task	Contributions
Comprehensive and reliable crowd assessment algorithms. **Joglekar et. al.** *ICDE 2015*	Micro-task	Focuses on "conciseness". Proposed solutions such as when not all workers have attempted every task, tasks have non-Boolean responses, workers have different biases for positive and negative tasks.
Measuring crowd truth: disagreement metrics combined with worker behavior filters. **Soberón et. al.** *CrowdSem 2013*	Micro-task	Identifies quality or spam workers as those who systematically disagree both with the majority and with the rest of co-workers.
Worker Skill Estimation in Team-Based Tasks. **Rahman et al.** *VLDB 2015*	Collaborative Tasks	Estimates worker skills in collaborative tasks for different skill aggregation models.

Task-Specific: Feedback

paper	Contributions
Web Workers, Unite! Addressing Challenges of Online Laborers. ***B. Bederson, A. J. Quinn.*** *CHI '11 Extended Abstracts*	*Immediate quality feedback:* CrowdFlower displays a panel above the task with the worker's estimated accuracy so far. *Longer-term feedback:* Send email to workers who submit 10 HITs below a quality threshold. Block after 20 HITs. *Disclose payment terms:* Turkers indicated that 24-48 hours as the max acceptable delay between submitting work and receiving payment. *Provide grievance process:* Provide a means for workers to request a review of work that was rejected. Requestors should explain why work was rejected.

Task-Specific: Incentives

paper	Contributions
Designing Incentives for Inexpert Human Raters. ***A. D. Shaw, J. J. Horton, D. L. Chen.*** *CSCW 2011*	Found that asking subjects to consider answers of others (social incentives) produces more accurate results on a content analysis task. Penalizes worker if disagreement with majority. Bonuses for surprising answers (financial incentives).
Finish Them!: Pricing Algorithms for Human Computation. ***Y. Gao, A. G. Parameswaran.*** *PVLDB 2014*	Algorithms for dynamic pricing to meet (a) user-specified deadline while minimizing total monetary cost, or (b) user-specified budget constraint while minimizing total elapsed time. Markov Decision Processes provide a 30% cost reduction.
The Labor Economics of Paid Crowdsourcing. ***J. J. Horton, L. B. Chilton.*** *EC 2010*	Designers should give people natural targets that will increase output, though they should also consider that such schemes might seem manipulative and unethical and could backfire.
Incentivizing High Quality Crowdwork. ***A. Slivkins, S. Suri, J. Vaughan.*** *WWW 2015*	Focus on performance-based payments (PBP). Tests effect of varying quality threshold in order for workers to receive a bonus and effect of varying the bonus amount. Recommends to first run a pilot experiment to determine whether a task is effort-responsive, i.e., correlation between time spent and quality. If additional time spent leads to much higher quality, PBPs are recommended.

Workers and Tasks: Motivation

paper	Contributions
And Now for Something Completely Different: Improving Crowdsourcing Workflows with Micro-Diversions. ***P. Dai, J. M. Rzeszotarski, P. Paritosh, E. H. Chi.*** *CSCW 2015* *Inserting Micro-Breaks into Crowdsourcing Workflows.* ***J. M. Rzeszotarski, E. H. Chi, P. Paritosh, P. Dai.*** *AAAI 2013*	Use diversions to improve workers' experience. Diversions = feedback encouraging text and a board showing how well worker is doing. Or, a funny video for 30 seconds. Found that diversions improve worker retention rate while retaining the same work quality. Suggests that automatic task assignment could maximize worker productivity and engagement.
Motivation through the design of work: Test of a theory. ***J. Hackman and G. R. Oldham.*** *Organizational behavior and human performance 1976.*	Modeled *extrinsic* and *intrinsic* motivation as the problem of job redesign. Tested the model against worker psychological state in a physical workplace.
An assessment of intrinsic and extrinsic motivation on task performance in crowdsourcing markets. ***J. Rogstadius, V. Kostakos, A. Kittur, B. Smus, J. Laredo, M. Vukovic****. ICWSM 2011*	Experimental study suggests that intrinsic motivation improves quality. Empirically demonstrates that increasing levels of payment increases work output regardless of intrinsic value. Finds a synergistic interaction between intrinsic and extrinsic motivators.

Motivation in the Social Sciences

Motivation through the design of work: Test of a theory. ***J. Hackman and G. R. Oldham.*** *Organizational behavior and human performance, 1976*

$$\text{Motivating Potential Score (MPS)} = \left[\frac{\text{Skill Variety} + \text{Task Identity} + \text{Task Significance}}{3}\right] \times \text{Autonomy} \times \text{Feedback}$$

Motivation in the Social Sciences

Motivation through the design of work: Test of a theory. ***J. Hackman and G. R. Oldham.*** *Organizational behavior and human performance, 1976*

- 658 employees in 62 heterogeneous jobs (*white collar, blue collar, industry, services, urban and rural settings*) in 7 organizations.
- Goal: which Job Dimensions stimulate which Psychological States: *experienced meaningfulness of the work, experienced responsibility for the work outcomes, knowledge of the actual results of the work.*
- Proposed model good for job design, i.e.
  - in determining the potential of a job to engender motivation,
  - in identifying which jobs need improvement,
  - in assessing the readiness of employees to respond to a redesigned job

Intermediate Takeaways

- Worker-specific factors, *Skill, Reputation/Trust, Expected pay, Acceptance ratio,* are modeled, acquired, learned.
- Task-specific factors, *Expertise, Feedback, Incentives*, are leveraged for worker motivation.
- *How are human factors leveraged in task assignment and task completion?*

20

PART 2
Human Factors in Task Assignment and Task Completion

Self-appointment to tasks in AMT

amazonmechanical turk
Artificial Artificial Intelligence

Sign In

Your Account | HITs | Qualifications

190,184 HITs available now

All HITs | HITs Available To You | HITs Assigned To You

Find HITs containing that pay at least $ 0.00 for which you are qualified require Master Qualification GO

All HITs

1-10 of 1518 Results

Sort by: Expiration Date (soonest first) GO!

Show all details | Hide all details

1 2 3 4 5 Next Last

Items per Page: 10

CTRP: Type name, date and total of a receipt — View a HIT in this group

Requester: CopyText Inc. **HIT Expiration Date:** Jan 14, 2016 (9 minutes 46 seconds) **Reward:** $0.01
Time Allotted: 4 minutes

Improve a Transcript — View a HIT in this group

Requester: CastingWords **HIT Expiration Date:** Jan 14, 2016 (10 minutes 50 seconds) **Reward:** $0.19
Time Allotted: 8 hours

Identify groups announced from audio recording (Level 2) — View a HIT in this group

Requester: TestNotice **HIT Expiration Date:** Jan 14, 2016 (19 minutes 59 seconds) **Reward:** $0.25
Time Allotted: 10 minutes

Micro-Task Assignment (limited acknowledgement of Human Factors)

Human Factors	Characteristics	Papers
Skill/Worker accuracy, Cost	(a) Each worker has a known and possibly different skill and cost (b) Assignment problem for matching workers to tasks to maximize quality and minimize/satisfy cost	(a) Online Task Assignment in Crowdsourcing Markets. Chien- Ju-Ho et al. AAAI 2012 (b) Adaptive task assignment for crowdsourced classification. . Chien- Ju-Ho et. al. JMLR 2013 (c) Budget-optimal task allocation for reliable crowdsourcing systems. D. Karger et. al. Operations Research 2014 (d) Personalized task recommendation in crowdsourcing information systems—Current state of the art. D. Geiger. Decision Support Systems, 2014
Skill/Worker accuracy, Cost	(a) Workers are indistinguishable - a generic error/skill/cost known for all workers (b) Find the best subset of tasks for workers to maximize quality under cost budget.	(a) Filtering: CrowdScreen: Algorithms for filtering data with humans. A. Parameshwaran et. Al. VLDB 2010 (b) Sorting and joins: So Who Won? Dynamic Max Discovery with the Crowd. S. Guo. SIGMOD 2012 (c) Human-powered sorts and joins. A. Marcus et. al. VLDB 2011 (d) Deduplication and clustering: Crowder: Crowdsourcing entity resolution. J. Wang et al. VLDB 2012 (e) Categorization: Crowdsourcing subjective image quality evaluation. R. Rebeiro et al. Int Conference on Image Processing 2011

Collaborative Task Assignment

with H. Rahman, S. Thirumuruganathan, G. Das. VLDBJ 2015.

- **Input**: tasks to complete, human workers
- **Output**: completed tasks

- Each task has Expertise/Quality/Cost requirements
  - *English comprehension* to check translation *correctness*

- Human factors per worker : *Skill, Expected wage, Acceptance ratio, Max/Min number of tasks*

- **Problem**: form one team per task

Example: Maximize outcome quality under task-centric and worker-centric constraints

objective: maximize aggregated v_t

$$\text{Maximize } \mathcal{V} = \Sigma_{\forall t \in T} v_t$$

aggregated worker skills and wages

task quality constraint

$$v_t = \begin{cases} W_1 \times \Sigma_{\forall j \in \{1..m\}} q_{t_j} + W_2 \times (1 - \frac{w_t}{W_t}) & \text{if } q_{t_j} \geq Q_{t_j} \\ & \wedge\ w_t \leq W_t \\ 0 & \text{if } q_{t_j} < Q_{t_j} \\ & \vee\ w_t > W_t \end{cases}$$

task budget

where $W_1, W_2 \geq 0$ and $W_1 + W_2 = 1$.

Solution Overview

Task Assignment: NP-hard (reduction using Multiple-Knapsack Problem)

- Our approach: Offline indexing for a workload of tasks
  - Assume a given set of tasks
  - Optimize for that set
- Solution:
  - A greedy deterministic algorithm with a 1-1/e approximation factor when sub-modular and monotonic
  - A greedy randomized algorithm with a 2/5 approximation factor when sub-modular

Task Completion: Online index maintenance when tasks arrive

- How to assign workers to new tasks?
- How to replace a worker who is not available for a task?

- Solution:
  - Solve the offline problem marginally
  - Efficient heuristics

Quality Experiments

- **Phase 1**: 8 multi-choice questions per task, to assess skills
- **Phase 2**: Collaborative Document Editing task
  - 20 workers asked to produce reports on 5 different topics:
    1) *Political unrest in Egypt,*
    2) *NSA document leakage,*
    3) *Playstation games,*
    4) *All electric cars*
    5) *Global warming*
- **Phase 3**: Completed tasks evaluated by crowd workers
  - 150 AMT workers evaluated Completeness, Grammar, Neutrality, Clarity, Timeliness, Added-Value

Outcome Quality Assessment

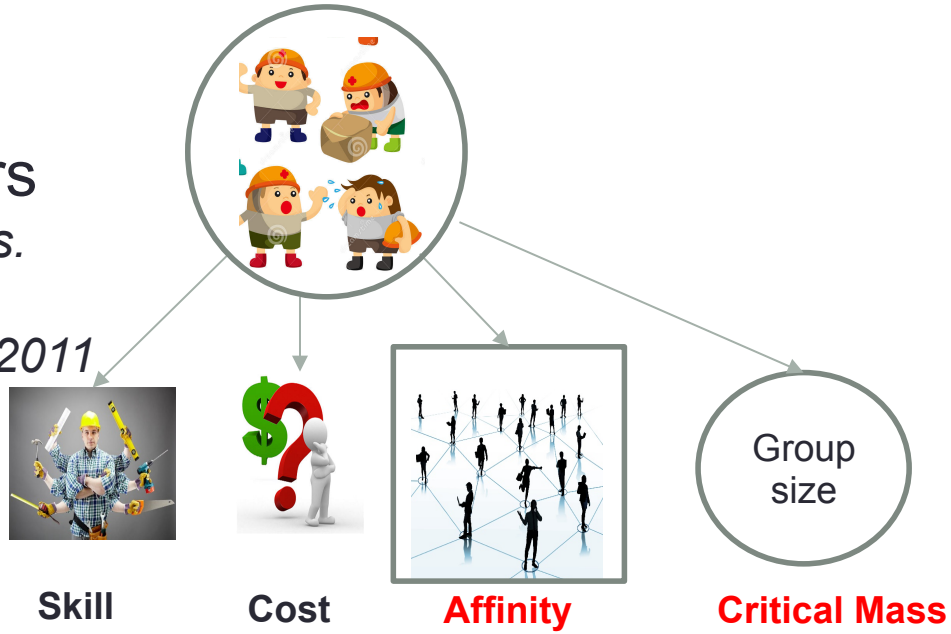
Average Rating							
Task	Algorithm	Completeness	Grammar	Neutrality	Clarity	Timeliness	Added-value
Egypt political unrest	SmartCrowd	4.5	4.2	4.0	4.2	4.1	4.0
	Online-greedy	3.3	3.4	3.3	3.0	3.4	3.1
	Benchmark	3.1	3.2	3.3	3.1	3.2	2.9
NSA document leakage	SmartCrowd	4.5	4.7	4.3	3.9	4.1	4.1
	Online-greedy	3.2	3.4	3.3	3.3	3.0	2.9
	Benchmark	3.3	3.3	3.4	2.9	2.9	3.4
PS Games	SmartCrowd	4.3	4.2	4.0	4.1	4.1	4.2
	Online-greedy	3.2	3.3	3.3	3.1	3.0	2.9
	Benchmark	3.0	3.2	3.1	2.8	2.9	2.9
All electric cars	SmartCrowd	4.2	4.2	4.1	4.4	4.0	4.1
	Online-greedy	3.0	3.1	3.3	3.0	2.9	2.8
	Benchmark	2.9	2.6	2.6	3.0	2.8	2.3
Global warming	SmartCrowd	4.2	4.3	4.5	4.2	4.1	3.7
	Online-greedy	3.0	3.2	3.1	3.4	3.3	3.3
	Benchmark	2.9	2.9	3.1	3.2	2.9	2.7

Bad news

- In some cases despite assigning tasks to highly qualified workers, outcome quality was low
  - Conflicting opinions
  - Edit wars
  - Proliferation of edits

- Group-aware human factors

  Synergetic effects in working teams.
  G. Hertel and G. Hertel.
  Journal of Managerial Psychology 2011



Interaction Model of Affinity

- Type Indicator: MBTI. ***Myers and Briggs****. Consulting Psychologists Press 1988*
- Are two heads better than one? Crowdsourced translation via a two-step collaboration of non-professional translators and editors. ***R. Yan et. al.*** *ACL 2014*

- *Pairwise*: using socio-demographic attributes, such as region, age, or psychological characteristics
- *Intra-group distance*: e.g., edit wars

$$DiaDist(\mathcal{G}) = Max_{\forall u_i, u_j \in \mathcal{G}} dist(u_i, u_j)$$

Critical Mass

-Managing research quality: critical mass and optimal academic research group size. ***R. Kenna et. al****. IMA Journal of Management Mathematics 2012*

- A group size beyond which collaboration effectiveness diminishes
- A large group (e.g., beyond size 10) is less effective

- Objective:

$$\text{Minimize } \{DiaDist(\mathcal{G}) + \sum_{\forall G_i, G_j \in \mathcal{G}} SumInterDist(G_i, G_j)\}$$

- Constraints:

$$\Sigma_{\forall u_i \in \mathcal{G}} u_{d_i} \geq Q_i \quad \forall d_i$$

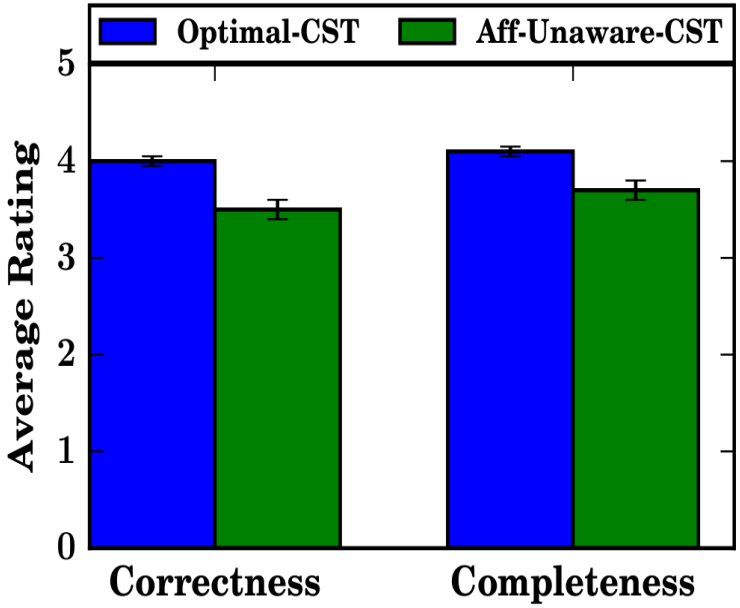
$$\Sigma_{\forall u \in \mathcal{G}} w_u \leq C$$

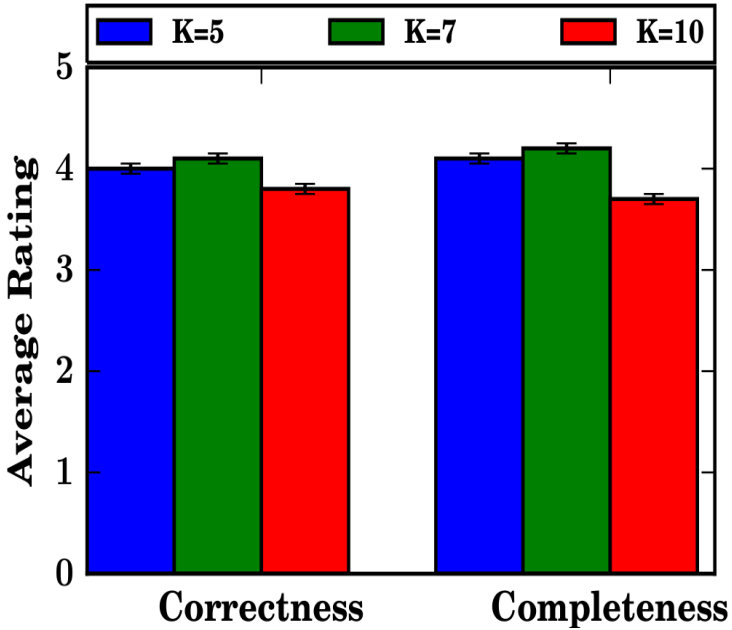
$$|G_i| \leq K \quad \forall i = \{1, 2, \ldots, x\}$$

Experiments with Affinity and Critical Mass

with H. Rahman, S. Thirumuruganathan, G. Das. ICDM 2015.

- Translation task with 120 AMT workers
- Region-based and age/gender-based affinities
- Evaluation based on Word Error Rate
  - Higher affinity impacts positively quality
  - A group beyond size 10 is less effective
  - Region-based more effective than age/ gender-based





Intermediate Takeaways

- Worker-specific factors are leveraged today in task assignment and completion
  - Micro-tasks: Skill, Reputation/Trust, Expected pay, Acceptance ratio
  - Collaborative tasks: Affinity, Critical mass, Interaction model
- Worker affinity combined with critical mass affect outcome quality in collaborative editing
- *How is evaluation actually conducted?*

33

PART 3
Evaluation and Future Directions

Evaluation points

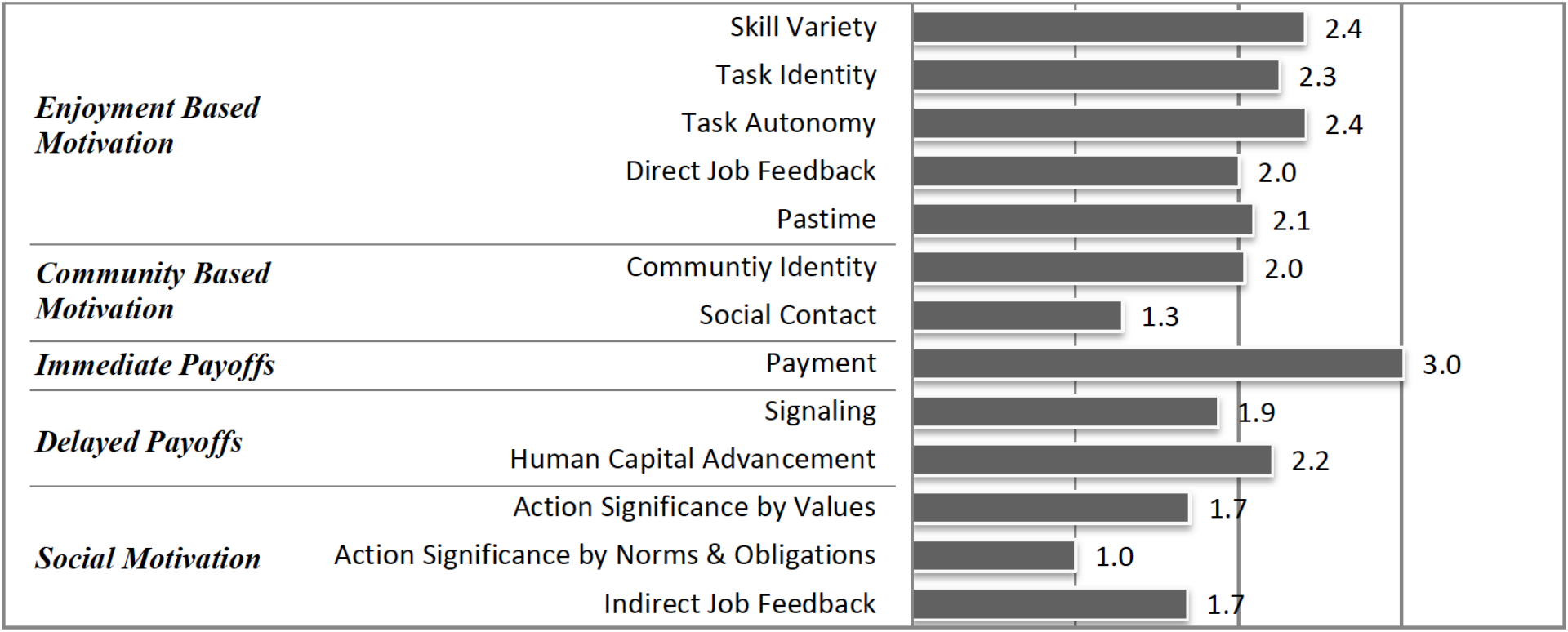
- **Goal**: evaluate the performance of a crowdsourcing platform
  - For different task assignment strategies
  - For different task completion environments
  - For both
- Evaluation protocols
  - Explicit or implicit

Evaluation Protocols

- Explicit: use questionnaires to measure *worker-centric dimensions*
  - before completing tasks: workers' expected performance
  - after: workers' overall experience with platform
  - before and after: workers' psychological state
- Implicit: observe workers during task completion and measure *requester-centric and worker-centric dimensions*
  - workers' overall experience with platform
  - worker retention
  - task throughput
  - payment

Explicit Worker-Centric Measures

More than fun and money. worker motivation in crowdsourcing-a study on mechanical turk. ***N. Kaufmann, T. Schulze, and D. Veit****. AMCIS 2011*



Does Money Matter? Motivationa l Factors for Participation in Paid and Non-Profit Crowdsourcing Communities. ***D. Pilz and H. Gewald.*** *International Conference on Wirtschaftsinformatik 2013.*

Two important observations on evaluation

1. Does explicit evaluation quantify outcome quality?
2. How is evaluation result accounted for in improving performance?

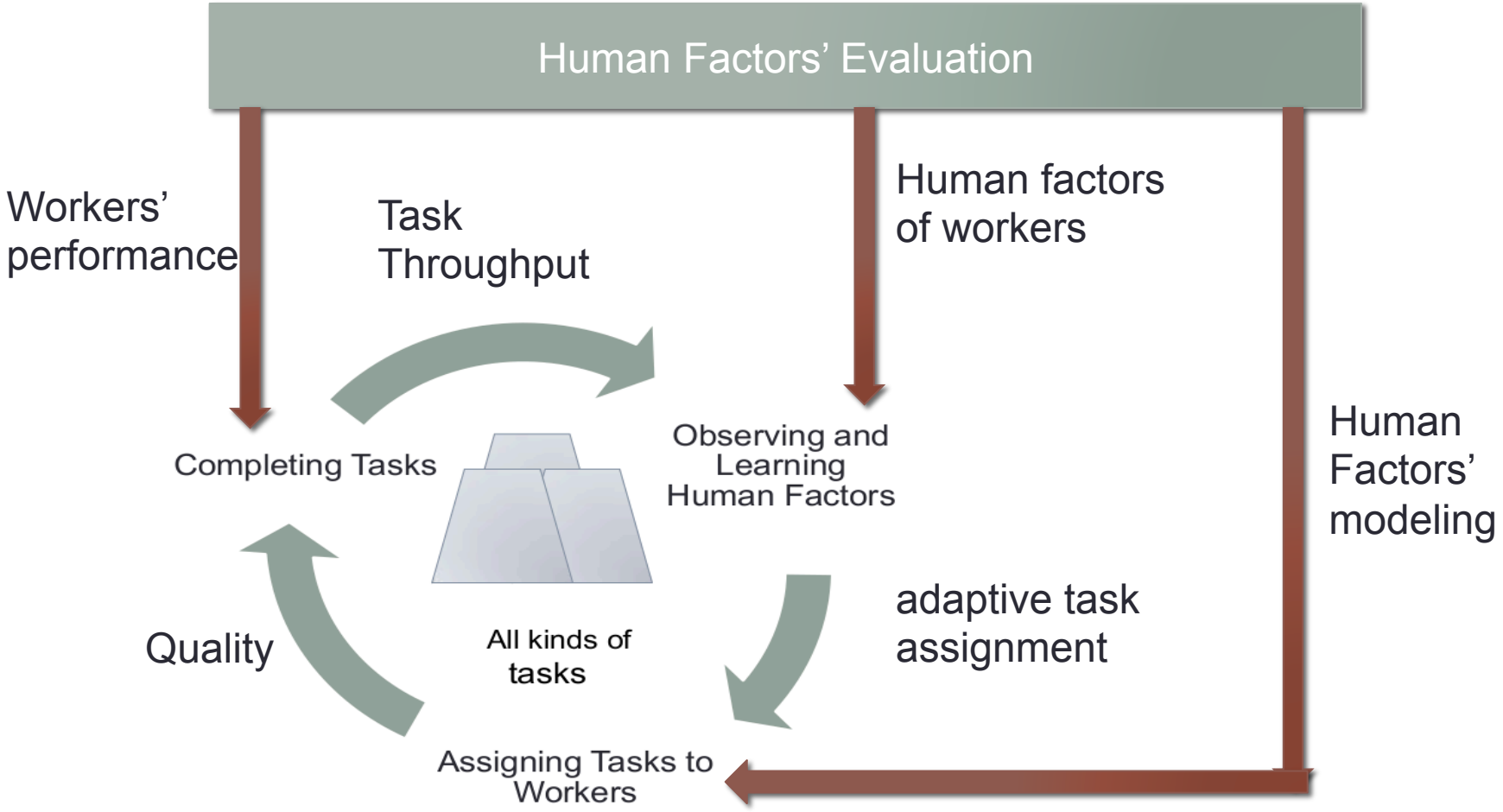
Does explicit evaluation quantify outcome quality?

- Longer worker retention does not necessarily imply higher outcome quality
- The same observation could be made for all other measures (task throughput, payment, workers' psychological state)
- Similarly for collaborative tasks
  - How does group size affect outcome quality of collaborative tasks?
  - How do interaction models affect quality?

Evaluating Outcome Quality

- Unless evaluated explicitly, other measures do not guarantee high outcome quality
  - Quality evaluation against a ground-truth
    - As in IR
  - Crowdsourced quality evaluation
    - E.g., ask a set of workers to evaluate the accuracy of a text translation and the quality of the output text

How is evaluation result accounted for in improving performance?



Motivation-Aware Task Assignment

with J. Pilourdault, D. Lee (work in progress)

- Challenge:
  - Workers' motivation evolves
- Approach:
  - Observe workers and adaptively assign tasks that maximize their motivation
- Different from entertaining workers during task completion (*Dai et al, CSCW 2015*), or incentivizing workers with gradually increasing pay (*Gao et al, PVLDB 2014*)

Motivation-Aware Task Assignment

with J. Pilourdault, D. Lee (work in progress)

- What is motivation?
  - 13 factors in the literature
  - Balance between intrinsic factor: task diversity and extrinsic factor: task payment

$$motiv_w^i(\mathcal{T}_w(i)) =$$

$$2\alpha_w^i \times TD(\mathcal{T}_w(i)) + (|\mathcal{T}_w(i)| - 1)(1 - \alpha_w^i) \times TP(\mathcal{T}_w(i))$$

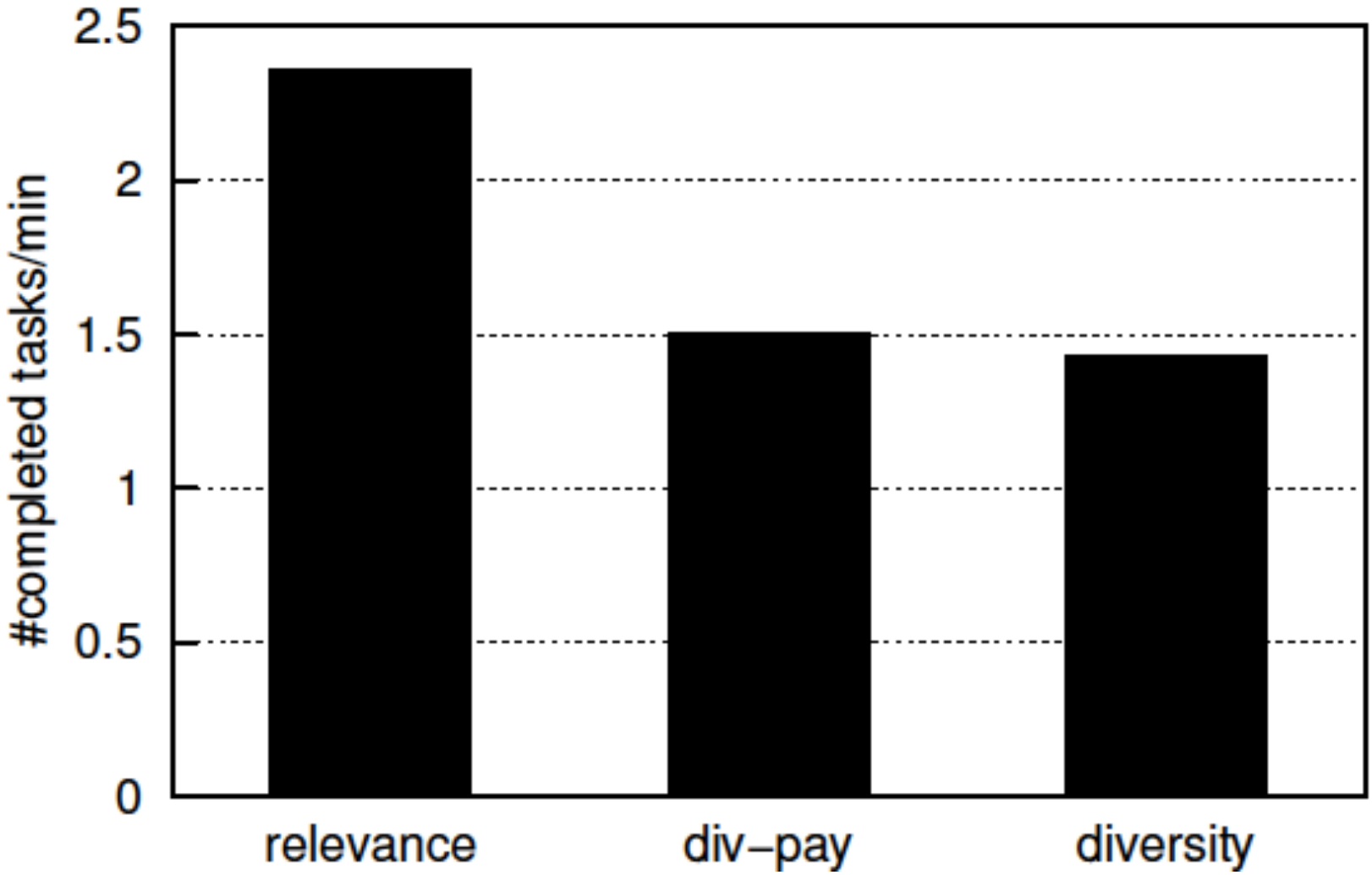
Motivation-Aware Task Assignment (`Mata`) *At each iteration i, and for each worker $w \in \mathcal{W}$, choose a subset of tasks $\mathcal{T}_w(i) \subseteq \mathcal{T}$ such that:*

$$\begin{aligned} \textbf{max} \quad & motiv_w^i(\mathcal{T}_w(i)) \\ \textit{such that} \quad & \forall t \in \mathcal{T}_w(i) \ matches(w,t) \quad (C_1) \\ & \sum_{t \in \mathcal{T}_w(i)} |\mathcal{T}_w(i)| \leq X_{max} \quad (C_2) \end{aligned}$$

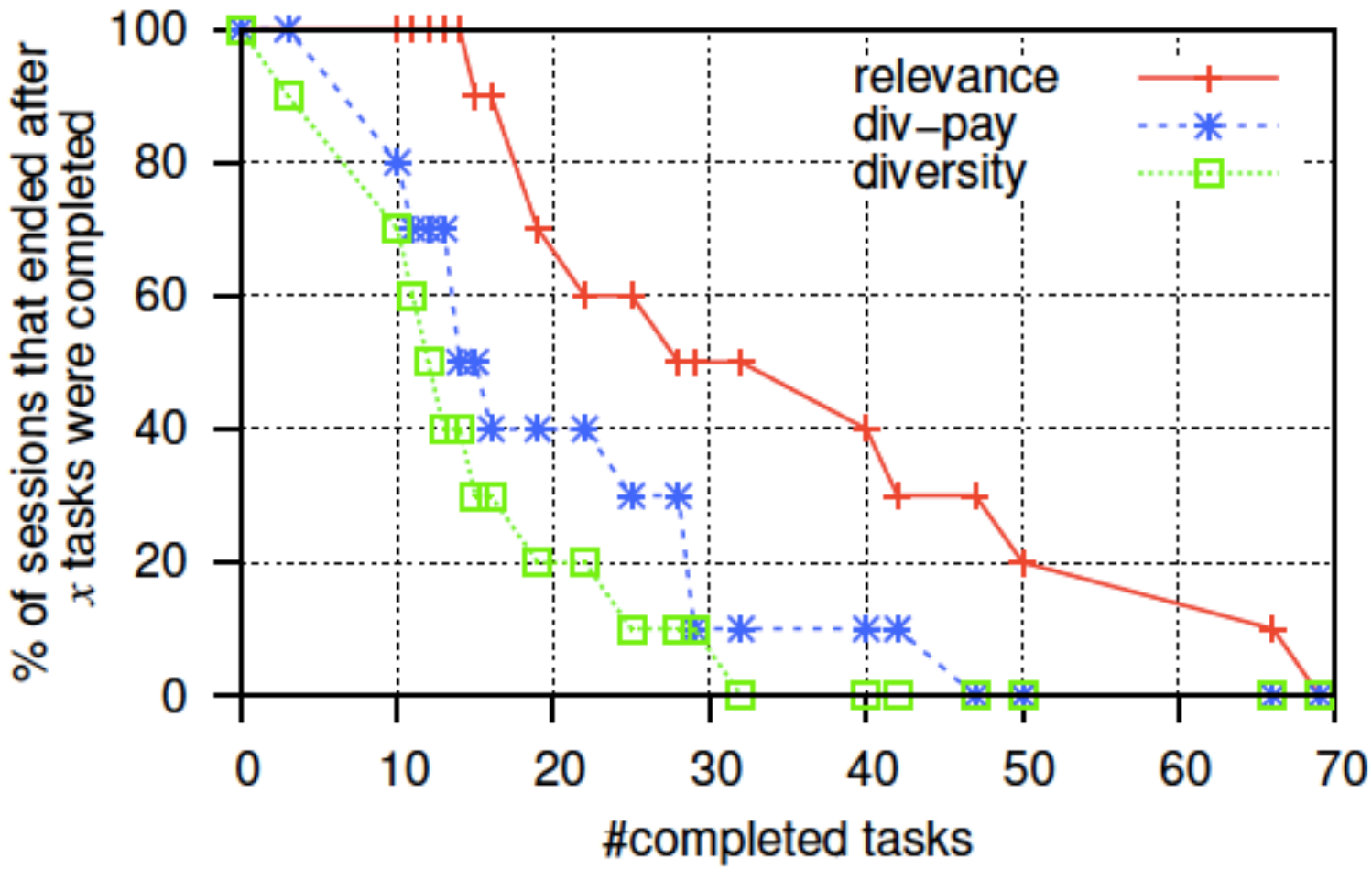
Motivation-Aware Task Assignment Performance

- 158,018 tasks from CrowdFlower in 22 kinds
- 23 workers in AMT
- 3 task assignment strategies:
  - Tasks matching a worker's profile: RELEVANCE
  - Diverse tasks matching a worker's profile: DIVERSITY
  - Tasks achieving a balance between diversity and payment: DIV-PAY
- Explore impact on performance

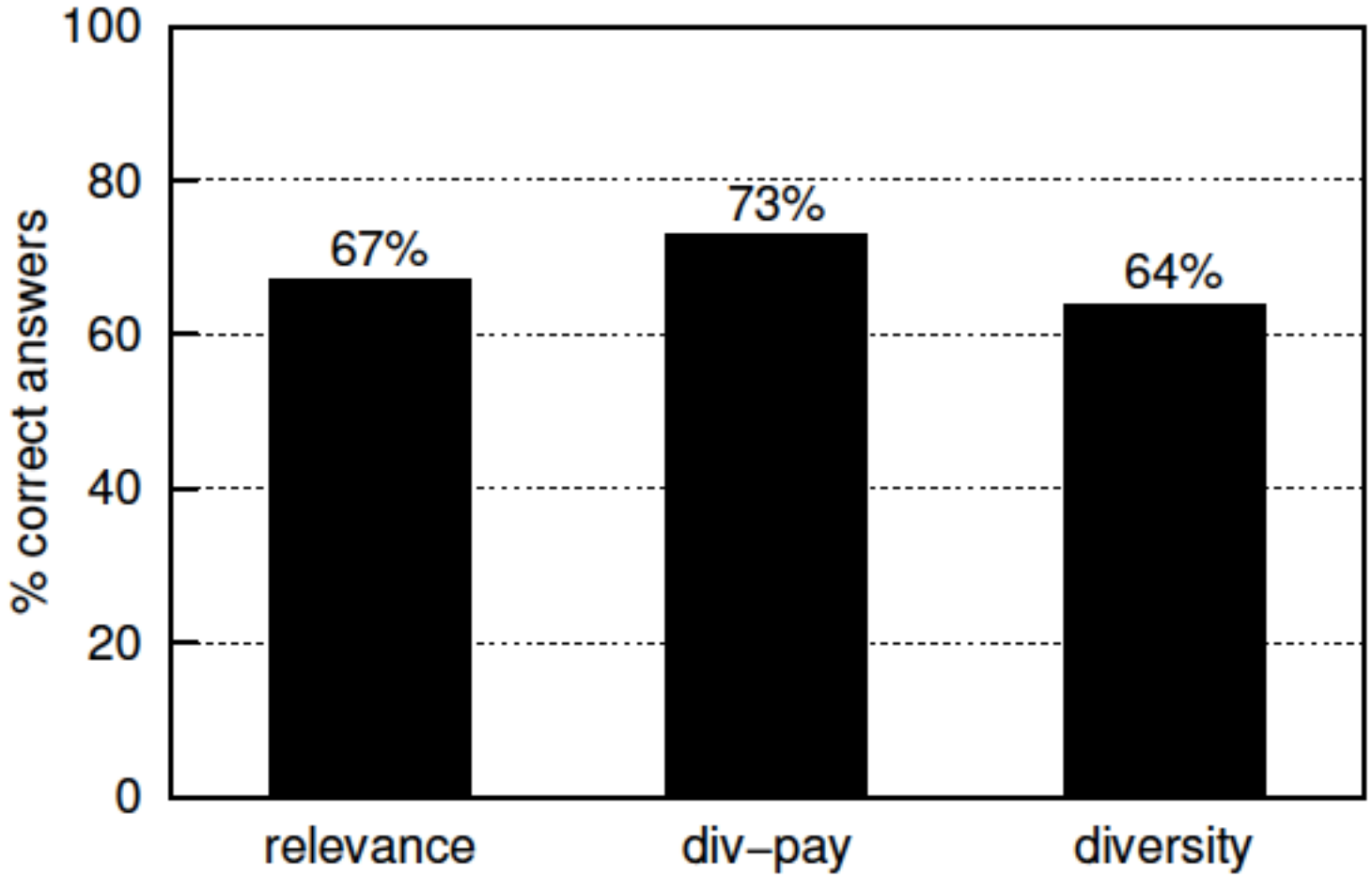
Bad news again? Task Throughput



Bad news again? Worker Retention



On the importance of measuring quality



Summary and Future Directions

49

Summary

- Human factors are an essential component of crowdsourcing platforms.
- Their importance differs for different kinds of tasks.
- They affect task assignment and task completion.
- They affect the way we conduct experimental evaluation.
- Their evolving nature requires to re-think how they are integrated into crowdsourcing processes.

- But we need to be even more careful about how evaluation is conducted
  - Result aggregation
  - Skill aggregation
  - Repeatability

Result Aggregation for Micro-Tasks

- Voting-based Methods
  - Majority Voting
  - Cumulative Weighted Voting
- Aggregative Methods
  - T-Norm
  - Weighted Sum
- Non-Aggregative Methods
  - Lexicographic
  - Consensus-based

Open question: revisit outcome quality with different aggregations of individual worker contributions

51

Skill Aggregation for Collaborative Tasks

Common functions:

- *Maximum*: e.g., collaborative editing $$\mathbf{q}_\ell = \max_j \mathbf{p}_\ell^j$$
- *Minimum*: e.g., fansubbing $$\mathbf{q}_\ell = \min\{1, \sum_{j=1}^{|Q|} \mathbf{p}_\ell^j\})$$
- *Multiplicative*: e.g, citizen science $$\mathbf{q}_\ell = (1 - \prod_{j=1}^{|Q|}(1 - \mathbf{p}_\ell^j))$$

Open question: revisit task assignment with different skill aggregation functions

Repeatability

- Experiment repeatability is complex
  - What is statistical significance in crowdsourcing?
- Different times of day attract very different workers
  - Workers with different socio-demographics
  - Workers in different time zones
- Different platforms attract different workers

pybossa

foulefactory

amazonmechanical turk
Artificial Artificial Intelligence
beta

CrowdFlower

Crowd4U

53

Acknowledgements (for sharing original slides and for providing feeback on earlier versions of tutorial)

- Laks Lakshmanan (UBC)
- Min Xie (Walmart Labs)
- Carlos Castillo (Eurecat)
- Aris Gionis (Aalto University)
- Saravanan Thirumuruganathan (QCRI)
- Habibur Rahman (UT Austin)

54

THE END